

CASSANDRA PHILOGENE

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EDUCATION

Concordia University

Bachelor of Engineering in Electrical Engineering

Montreal, QC

Winter 2024 – 2028 Expected

TECHNICAL SKILLS

Electronics Repair and Testing: troubleshooting, diagnostics, soldering, multimeter testing, oscilloscope operation, voltage analysis, component replacement, hardware assembly

Electrical Hardware: wiring and harnessing, PCB design, power distribution, sensor interfacing, electromechanical assemblies, circuit simulation, schematic and BOM review

Programming: C, C++, Python, VHDL

Embedded Systems/Robotics: Arduino, ROS2, motor control, autonomous systems, sensor integration

Tools: LTspice, MATLAB, EasyEDA, Altium Designer, Fusion 360, Git/GitHub, Linux, Vivado, ModelSim, AutoCAD Basic

Languages: French Native, English Fluent, Spanish Beginner

EXPERIENCE

FUNHUB Internship

May 2026 – Aug. 2026

Technician in Machine Repair and Maintenance

Montreal, QC

- Performed hands-on diagnostics, troubleshooting, repairs, and preventative maintenance on arcade and entertainment equipment to reduce downtime and support daily operations
- Identified electrical, mechanical, and system-level faults using a methodical troubleshooting approach
- Performed equipment inspections, functionality checks, multimeter testing, component replacements, and maintenance support across multiple arcade systems
- Worked with sensors, wiring, power systems, electromechanical assemblies, and control hardware while following safety and operational procedures
- Maintained technical records and clearly communicated equipment status, repair findings, and maintenance requirements to supervisors and team members

ENGINEERING PROJECTS

Programmable Synchronous Buck Converter PCB

Dec. 2025 – May 2026

SAE Aero Design

Montreal, QC

- Designed and implemented a synchronous buck converter PCB based on the LTC3892 controller to step down a 20 V input to selectable regulated outputs of 4.8 V, 6.0 V, 7.4 V, or 8.4 V
- Performed component sizing calculations and selected MOSFETs, inductors, capacitors, and protection circuitry based on electrical and efficiency requirements
- Created the complete schematic, Bill of Materials, and PCB layout while considering manufacturability, component availability, and system reliability
- Validated voltage regulation, transient response, switching behavior, output ripple, and converter stability through LTspice simulations
- Calculated copper trace widths based on current-carrying requirements, power dissipation, and thermal performance constraints
- Applied PCB design practices including grounding strategy, decoupling capacitor placement, current-loop minimization, and EMI-conscious routing
- Completed PCB routing, design verification, and manufacturing preparation while checking the design against electrical and thermal constraints

TECHNICAL ACTIVITIES

SAE Aero Design Competition – Systems Team

Jun. 2025 – Present

Concordia University Student Design Team

Montreal, QC

- Assisted in the assembly and integration of aircraft electrical systems including brushless motors, ESCs, BECs, receivers, gyroscopes, batteries, and control hardware
- Designed and assembled wiring harnesses, connectors, and power distribution systems for aircraft subsystem integration
- Supported testing, troubleshooting, and validation activities to verify communication, subsystem functionality, and overall system readiness
- Applied safe LiPo battery handling, charging, and operational procedures during testing and competition preparation